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Harnessing γδ T cells for B7-H3-targeting CAR therapy to enhance anti-tumor therapy in glioblastoma

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Abstract

Background Glioblastoma (GBM) remains largely incurable, in part because the highly immunosuppressive tumor microenvironment (TME) poses a major barrier to conventional therapies, including to chimeric antigen receptor (CAR)-T cell therapy. Compared with conventional αβ T cells, γδ T cells bridge innate and adaptive immunity, amplifying anti-tumor responses through cytokine secretion and cross-talk with other immune populations, so arming γδ T cells with a B7–H3-targeting CAR could combine their inherent tumor-homing and immunomodulatory capabilities with precise antigen-specific cytotoxicity for therapeutic synergy.

Methods CD276 (B7–H3) expression and prognostic relevance were analyzed using bulk and single-cell RNA-seq data from TCGA and CGGA cohorts. Human γδ T cells and conventional αβ T cells were expanded from healthy donor PBMCs and engineered with a second-generation B7–H3 CAR. CAR expression and phenotype were assessed by flow cytometry. Antitumor activity against B7–H3^{+} glioma cell lines (U87, U251) was evaluated using cytotoxicity and cytokine-release assays. Therapeutic efficacy was tested in NSG mice bearing glioma xenografts, with tumor growth, survival, tumor infiltration, apoptosis, and checkpoint expression assessed by immunofluorescence.

Results B7–H3 CAR-γδ T cells exhibited superior antitumor functionality compared with B7–H3 CAR-αβ T cells in GBM models. In vitro, B7–H3 CAR-γδ T cells displayed enhanced, sustained cytotoxicity against GBM cell lines and secreted significantly higher levels of key effector cytokines (IFN-γ, TNF-α) than B7–H3 CAR-αβ T cells, indicating a polyfunctional and exhaustion-resistant phenotype. In GBM mouse models, a single dose of B7–H3 CAR-γδ T cells mediated robust tumor control and significantly prolonged survival. Mechanistic studies traced this superior efficacy to enhanced tumor infiltration and more potent induction of tumor cell apoptosis.

Conclusions B7–H3 CAR-γδ T cells therefore represent a promising new approach to GBM immunotherapy which, through ongoing platform optimization and clinical exploration, could offer new therapeutic hope to patients with glioma.

Keywords B7-H3, CAR-γδ T cells, Glioblastoma, Immunotherapy, γδ T cell engineering

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Introduction

Glioblastoma (GBM) is the most common and lethal primary malignant tumor of the central nervous system [1]. Despite aggressive standard-of-care involving maximal safe resection, radiotherapy, and temozolomide chemotherapy, the prognosis remains dismal, with a median survival of approximately 15 months [2, 3]. The highly infiltrative nature of GBM, coupled with its profoundly immunosuppressive tumor microenvironment (TME), render most conventional and targeted therapies largely palliative, highlighting an urgent need for new therapeutic strategies [4, 5].

While immunotherapy approaches, particularly chimeric antigen receptor (CAR) T-cell therapy, have revolutionized the management of hematological malignancies, there are significant barriers to their application to solid tumors such as GBM [6, 7]. These barriers include antigenic heterogeneity and loss, inefficient T-cell trafficking and infiltration into tumor sites, and, critically, the rapid induction of T-cell exhaustion and functional impairment within the suppressive TME [7, 8]. Consequently, overcoming these limitations requires a dual strategy: identifying optimal tumor-associated antigens while developing more resilient effector cell platforms capable of sustained function within the GBM microenvironment [9].

Several antigens have been explored as CAR targets in GBM, but each has limitations. EGFRvIII, while tumor-specific, is only expressed in about a third of patients and exhibits significant intratumoral heterogeneity [10, 11]. IL13Rα2, though a validated target, is also variably expressed and can be downregulated under therapeutic pressure [12–14]. In contrast, B7–H3 (CD276) is characterized by its broad and homogeneous overexpression across the majority of GBMs and associated vasculature [15–17] and minimal expression in normal brain tissue [18–20]. B7–H3 is therefore an attractive, potentially widely applicable antigen for CAR-based therapy. Indeed, preclinical studies have shown robust antitumor activity of B7–H3 CAR T cells in GBM xenograft models [21], although efficacy may still be constrained by the immunosuppressive TME and by functional exhaustion driven by chronic antigen stimulation [22, 23]. Therefore, the overall performance of B7–H3 CAR therapy is likely to be influenced not only by target selection but also by the fitness of the effector cell platform. Conventional αβ T cells often show limited persistence and rapid functional impairment in solid tumors [24–26], driving exploration of alternative cellular carriers. γδ T cells are particularly appealing given their MHC-independent target recognition, intrinsic tumor tropism, and relative resistance to immunosuppressive cues, while also promoting broader immune activation [25, 27, 28]. We therefore hypothesized that pairing B7–H3 targeting with a γδ T-cell chassis would improve CAR-T trafficking, persistence, and functional durability in GBM. Specifically, we propose that B7–H3–specific CAR-γδ T cells will outperform conventional B7–H3 CAR-αβ T cells by combining antigen-directed cytotoxicity with the inherent resilience of γδ T cells [29].

To test this hypothesis, here we engineered identical second-generation B7–H3-specific CARs into both γδ T cells and conventional αβ T cells from matched healthy donors. Through comprehensive in vitro and in vivo analyses, we provide functional evidence that B7–H3 CAR-γδ T cells exhibit enhanced cytotoxicity, superior tumor infiltration, sustained polyfunctional cytokine responses, and a more exhaustion-resistant phenotype. Our findings establish B7–H3 CAR-γδ T cells as a potent new immunotherapeutic candidate for GBM and, more broadly, validate γδ T cells as an optimized cellular platform for next-generation CAR therapies against solid malignancies.

Materials and methods

Data collection

Bulk RNA-seq expression data and corresponding clinical information were obtained from three public sources. Data for 757 primary glioma tumors were obtained from The Cancer Genome Atlas (TCGA-GLIOMA) via the Genomic Data Commons (GDC) data portal (https://portal.gdc.cancer.gov/). CGGA_693 and CGGA_325 data were downloaded from the Chinese Glioma Genome Atlas (CGGA, https://www.cgga.org.cn/), which contained 422 and 229 primary glioma samples, respectively, for subsequent analysis [30].

Processed single-cell RNA-seq data and annotation information were obtained from the Cancer Single-cell Expression Map (CancerSCEM, https://ngdc.cncb.ac.cn/cancerscem/) [31].

Survival analysis

Patients from each cohort were stratified into CD276^{high} and CD276^{low} groups based on CD276 gene expression. The optimal cut point was determined using the 'surv_cutpoint' function in the R package 'survminer' (v0.4.9). Overall survival analysis was performed using the Kaplan-Meier method and the log-rank test.

Differential gene expression analysis

Differential gene expression analysis was performed on a STAR count matrix using R package 'DESeq2' (v1.42.1) after filtering out genes expressed in less than 10 samples [32]. Genes with an adjusted *p*-value (Benjamini-Hochberg procedure) less than 0.05 and an absolute log2 fold-change greater than 1 were considered differentially expressed.

Functional enrichment of CD276-correlated genes

Pearson's correlations for CD276 expression were calculated for each gene. Genes with correlation coefficients > 0.5 ($p < 6e\text{-}49$) were considered highly correlated with CD276 and were used for functional enrichment analysis using R package 'clusterProfiler' (v4.10.1) [33]. The 'HALLMARK' collection and 'GO:BP' sub-collection and any term related to glioma or glioblastoma in MSigDB were included [34, 35].

Analysis of tumor microenvironment components

Cell type deconvolution results of TCGA samples were downloaded from Kassandra [36]. Fractions of 14 indivisible cell types, except component 'Other', were extracted and re-normalized to sum to 1, representing the fractions of corresponding cell types in the TME.

Cell lines and culture

U87 human brain astroblastoma cells and U251 human glioma cells were purchased from Wuhan Procell Biotech Company (Wuhan, China). HEK293T human embryonic kidney cells and HT-1080 human colon cancer cells were obtained from Wuhan Procell Biotech company (Wuhan, China) for lentivirus preparation and titer assays.

U87 cells were cultured in MEM supplemented with 10% fetal bovine serum (FBS), 1% non-essential amino acids, and 1% penicillin/streptomycin (PS). U251 cells were cultured in DMEM/F12 medium supplemented with 10% FBS and 1% PS. HEK293T and HT-1080 cells were maintained in DMEM supplemented with 10% FBS and 1 mM Glutamax. All cell lines were validated to be *Mycoplasma* free using the Rapid Mycoplasma Test kit (Cellapybio, Beijing, China), following the manufacturer's instructions.

Lentivirus generation

The anti-B7–H3 CAR, featuring a CD8 transmembrane domain, 4-1BB co-stimulatory domain, and CD3ζ signaling domain (BB.z), was constructed as described [37]. T lentivirus was produced in HEK293T cells by co-transfecting the CAR plasmid with packaging plasmids (pLP1, pLP2, pVSV-G) at a 1:1:1:1 ratio using Lipofectamine 3000. Viral supernatants were collected, concentrated by ultracentrifugation, titrated on HT1080 cells, and stored at −80 °C

Expansion and transduction of γδ T cells

Primary human γδ T cells were expanded from healthy donor peripheral blood mononuclear cells (PBMCs) (Table S1). All donors provided written informed consent, and the study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Xuanwu Hospital, Capital Medical University (approval number: [2021]021). Briefly, PBMCs were isolated by Ficoll–Paque density gradient centrifugation. Cells were seeded at a density of 5×10^6 cells/mL in culture medium supplemented with 100 IU/mL IL-2, 50 ng/mL IL-15, 20 ng/mL TGF-β, and 2 μM zoledronic acid. The medium was refreshed every 2–3 days. On day 7, cells were transduced with lentivirus encoding the B7–H3specific CAR or a control construct on Retronectin-coated plates. The culture was maintained until day 21, with transduction efficiency and γδ T cell purity ($TCR\gamma\delta^+$) quantified by flow cytometry on day 14. Final products were harvested based on growth status, and viable cell counts were determined by trypan blue exclusion before use in experiments or cryopreservation.

Expansion and transduction of αβ T cells

Conventional αβ CAR-T cells were generated from the same healthy donor PBMCs. After isolation by Ficoll–Paque gradient centrifugation, $CD3^+$ T cells were enriched via positive selection and activated with CD3/CD28 magnetic beads (Gibco, Thermo Fisher Scientific) at a beadtocell ratio of 1:2. Activated T cells were seeded into ultralow attachment plates at 1.5×10^6 cells/mL in XVIVO 15 medium (Lonza, Basel, Switzerland) supplemented with 500 IU/mL IL-2 (XinLuoEr), 80 ng/mL IL7, 20 ng/mL IL15, and 20 ng/mL IL21 (all from Jin'an). Lentiviral transduction was performed on day 2–3 after activation. Cells were expanded for 10 days, after which transduction efficiency was evaluated by flow cytometry.

Flow cytometry

The following anti-human antibodies were procured from BioLegend (San Diego, CA, USA): CD3, CD4, CD8, B7–H3, PD-1, LAG3, TIM3, γδ TCR, NKp30, NKp44, NKp46, NKG2D (FITC, PE, PerCP, APC), and streptavidin-PE. Biotin-labeled Fab was acquired from Abcam (Cambridge, UK) for the quantification of CAR-positive T cells, with the presence of biotin-Fab detected through the addition of streptavidin-PE. Flow cytometry analysis was conducted using a FACSCalibur instrument (BD Biosciences, San Jose, CA), and the obtained data were analyzed using FlowJo v10.0.

In vitro cytotoxicity assay

CAR-T cells and glioma cell lines were co-cultured at various effector-to-tumor (E/T) ratios. In the transient cytotoxicity assay, the co-culture supernatant was collected from various E/T ratios after 20 hours of culture. In the multi-round cytotoxicity assay, effector cells were co-cultured with target cells at an E/T ratio of 4:1, with an equal number of target cells added every other day. Supernatant samples were collected every 48 hours, and the initial number of tumor cells was reintroduced into the co-culture system to re-challenge the CAR-T cells. Every 48 hours, the original number of tumor cells

(1×10^4 U87 or U251 cells per well) was reintroduced into the same culture. Effector cells were not replenished at any point during the assay. Supernatant was collected before each rechallenge to measure LDH and for cytotoxicity assays. Cytotoxicity was determined using a Non-Radioactive Cytotoxicity Assay Kit (Promega, Madison, WI, USA). Maximum LDH release control (high control) was obtained by lysing target cells alone with 1% Triton X-100 (final concentration). Specific lysis was calculated as: (Experimental release – Effector spontaneous release – Target spontaneous release)/(Maximum release – Target spontaneous release) × 100%.

Cytokine release assay

The cytotoxic cytokines granzyme, interferon (IFN)-γ, interleukin (IL)-2, and tumor necrosis factor (TNF)-α were quantified using ELISA kits in vitro following the manufacturer's instructions (Neobioscience, Shenzhen, China). Supernatants were collected from the samples and centrifuged at 800 × g for 15 minutes at 4 °C. A standard curve generated from serial dilutions of recombinant cytokine was included in every assay run to ensure accurate quantification across experiments and between experimental groups. ELISA data were acquired using a Varioskan Flash (Thermo Fisher Scientific, Waltham, MA, USA).

Immunofluorescence staining

Immunofluorescence staining was performed on fixed cell lines or tumor tissue sections. Cells on PDL-coated coverslips and 20 μm tumor sections were blocked with 5% donkey serum and incubated overnight at 4 °C with primary antibodies targeting EGFP, B7–H3, and active caspase-3 (1:50–1:200). After PBS washes, samples were incubated with Cy3- or Cy5-conjugated secondary antibodies, and nuclei were counterstained with DAPI. For quantification, five random non-overlapping fields per sample were imaged under identical settings, and positive cells were counted using ImageJ.

Animal studies

Six- to eight-week-old female NOD/SCID IL2Rγc NSG mice were obtained from Beijing Vitalstar Biotechnology Co., Ltd (Beijing, China, animal license numbers SYXK 2024–0025). Mice were subcutaneously injected with 1.0×10^6 U87/U251 tumor cells followed by 1.0×10^7 CAR-modified or control T cells i.v. at 7 and 10 days. Tumor volumes were measured every three days using calipers, and tumor volumes were calculated according to the formula volume = (diameter ^{L}×(diameter ^{S})^2)/2 (L, long diameter; S, short diameter). All mice were weighed every seven days, and mice were monitored daily for signs of clinical toxicity. Mice were maintained in cages of up to NSG mice under barrier conditions, and experiments were conducted using established protocols in a pathogen-free facility fully accredited by the Association for Assessment and Accreditation of Laboratory Animal Care. Mise were euthanized when they showed signs of excessive weight loss or other clinical signs of toxicity.

Statistical analysis

Data are presented as mean ± standard deviation (SD). Statistical analyses were performed using Prism (GraphPad Software, La Jolla, CA). Student's *t*-test was used as a two-sided paired test with 95% confidence intervals (CI) for comparisons between two groups. For comparisons involving three or more groups, two-way analysis of variance (ANOVA) was conducted with Dunnett's multiple comparisons test. Results with a *p*-value less than 0.05 were deemed statistically significant.

Results

B7–H3 and glioma prognosis

Many studies assessing B7–H3 protein and gene expression have consistently demonstrated its overexpression in gliomas, particularly high-grade gliomas such as glioblastoma, and expression levels correlate with advanced stage and a poorer prognosis [38–41]. To further investigate the clinical significance of CD276 (B7–H3) in glioma, we first performed survival analysis in multiple independent cohorts. Overall survival was significantly shorter for patients expressing high levels of *CD276* in the TCGA-GLIOMA, CGGA_693, and CGGA_325 datasets (all $p < 0.0001$), establishing its strong prognostic value (Fig. 1A). Transcriptomic profiling revealed that *CD276*^{high} tumors demonstrated coordinated upregulation of extracellular matrix components including *PCOLCE*, *CD248*, and *COL6A3*, while *CD276*-correlated genes were significantly enriched in mesenchymal transition, hypoxia response, and collagen organization pathways (Fig. 1B–D). While functional enrichment confirmed enhanced activity in IFN-γ response and antigen processing/presentation pathways, suggesting profound alterations in tumor-immune interactions (Fig. 1E), the coordinated increase in multiple immune checkpoint molecules, including *CD274* (PD-L1), *PDCD1* (PD-1) and *CTLA4* (Fig. 1F), and the differential abundance of cell types (Fig. 1G), especially enrichment of M2 macrophages, indicated an immunosuppressive microenvironment in CD276^{high} gliomas. Notably, single cell-resolution analysis revealed significantly increased *CD276* expression in malignant cells compared to oligodendrocytes ($p = 4.3e\text{-}14$), indicating tumor cell specificity (Fig. 1H). These integrated analyses establish CD276 as a central regulator linking glioma aggressiveness with immunosuppression, highlighting its dual role in promoting mesenchymal transition while

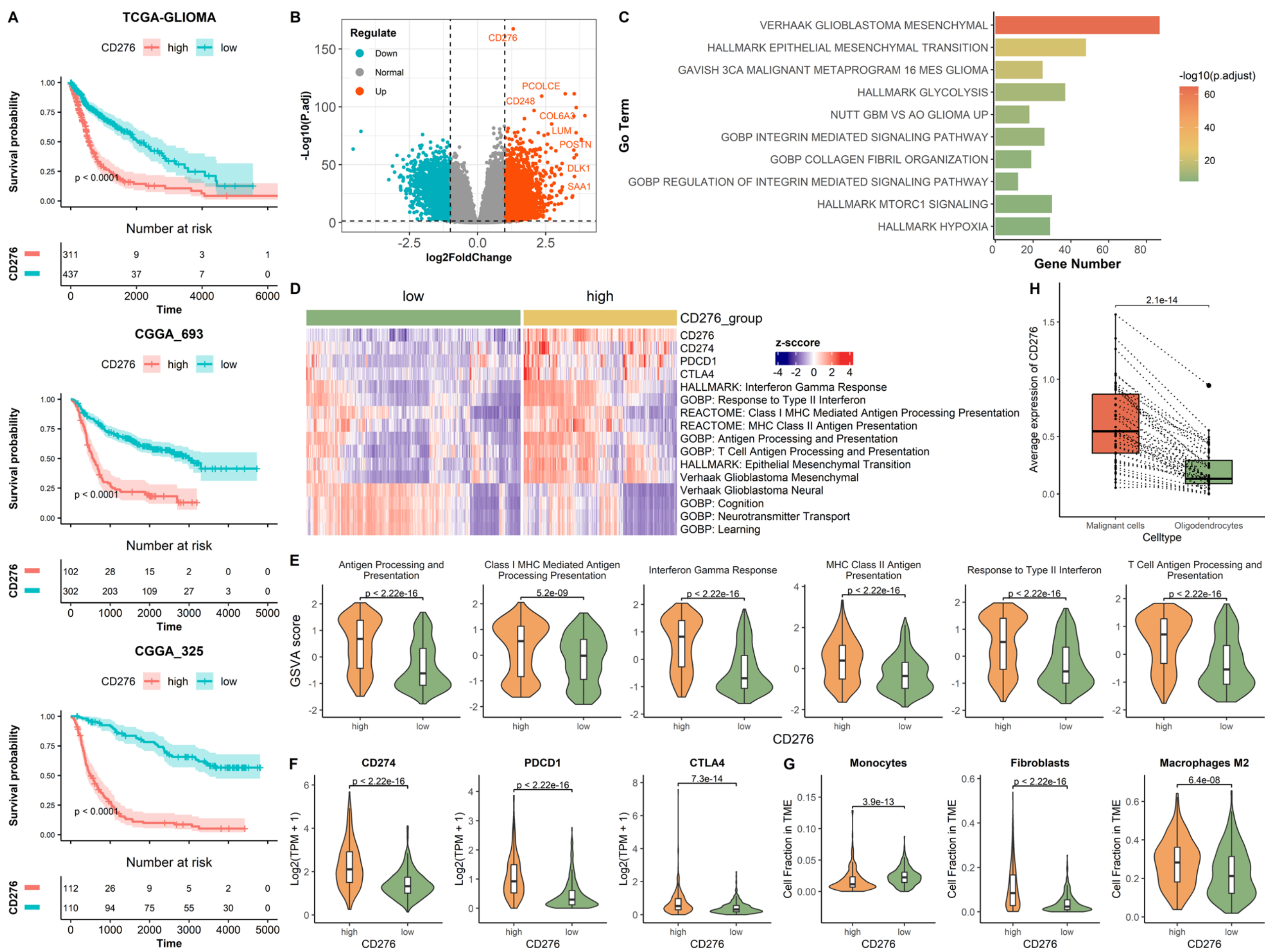

Fig. 1 *CD276* expression is associated with a poor prognosis and immune suppression in gliomas. This integrated analysis establishes CD276/B7–H3 as a key regulator linking glioma aggressiveness with an immunosuppressive microenvironment. (**A**) Kaplan-meier survival analysis showing overall survival of patients with glioma stratified by high and low *CD276* expression. (**B**) Volcano plot displaying differentially expressed genes between *CD276*^{high} and *CD276*^{low} gliomas. Selected genes are labeled. (**C**) Functional enrichment analysis of *CD276*-correlated genes. (**D**) Heat map of immune checkpoint gene expression and GSVA scores of selected pathways. Comparison of (**E**) GSVA scores of immune-related pathways, (**F**) Immune checkpoint gene expression, and (**G**) TME components between *CD276*^{high} and *CD276*^{low} groups. (**H**) Average expression of *CD276* in malignant cells compared with that in oligodendrocytes from the same sample

orchestrating an immune-evasive microenvironment. It can also serve as a target for glioma immunotherapy.

γδ T cells are cultivatable in vitro

γδ T cells recognize diverse, non-peptide antigens, so their expansion is more challenging than for αβ T cells, which can be activated with anti-CD3 antibodies. Furthermore, γδ T cells are highly heterogeneous, requiring specific, varied, and non-standardized cytokine signals for expansion [42]. The in vitro culture and expansion protocol for γδ T cells isolated from PBMCs is illustrated in Fig. 2A. The ratio of γδ to αβ T cells in the culture system was assessed at various time intervals during in vitro cultivation. Over time, the number of γδ T cells increased in the culture system and the number of αβ T cells decreased (Fig. 2B). After 21 days of culture, the proportion of γδ T cells, as well as the relevant surface markers, were assessed (Fig. 2C, D), confirming that it is possible to specifically amplify γδ T cells. The moderate expression of NKp46 and NKp30 indicated that the cultured γδ T cells may be capable of responding to external tumor signals, while the extremely low NKp44 expression indicated a non-activated state. γδ T cells are expected to exhibit potent anti-tumor capabilities upon further stimulation.

CAR structure and expression of CAR-positive T cells

To compare the anti-tumor efficacy of CAR-αβ T cells and CAR-γδ T cells, we first designed a CAR structure targeting B7–H3 (Fig. 3A) using two glioblastoma cell lines (U87 and U251) expressing high levels of B7–H3 (Fig. 3B). Expressing CAR on the surface of transduced αβ T cells and γδ T cells would yield B7–H3 CAR-αβ T cells and B7–H3 CAR-γδ T cells. To evaluate the feasibility

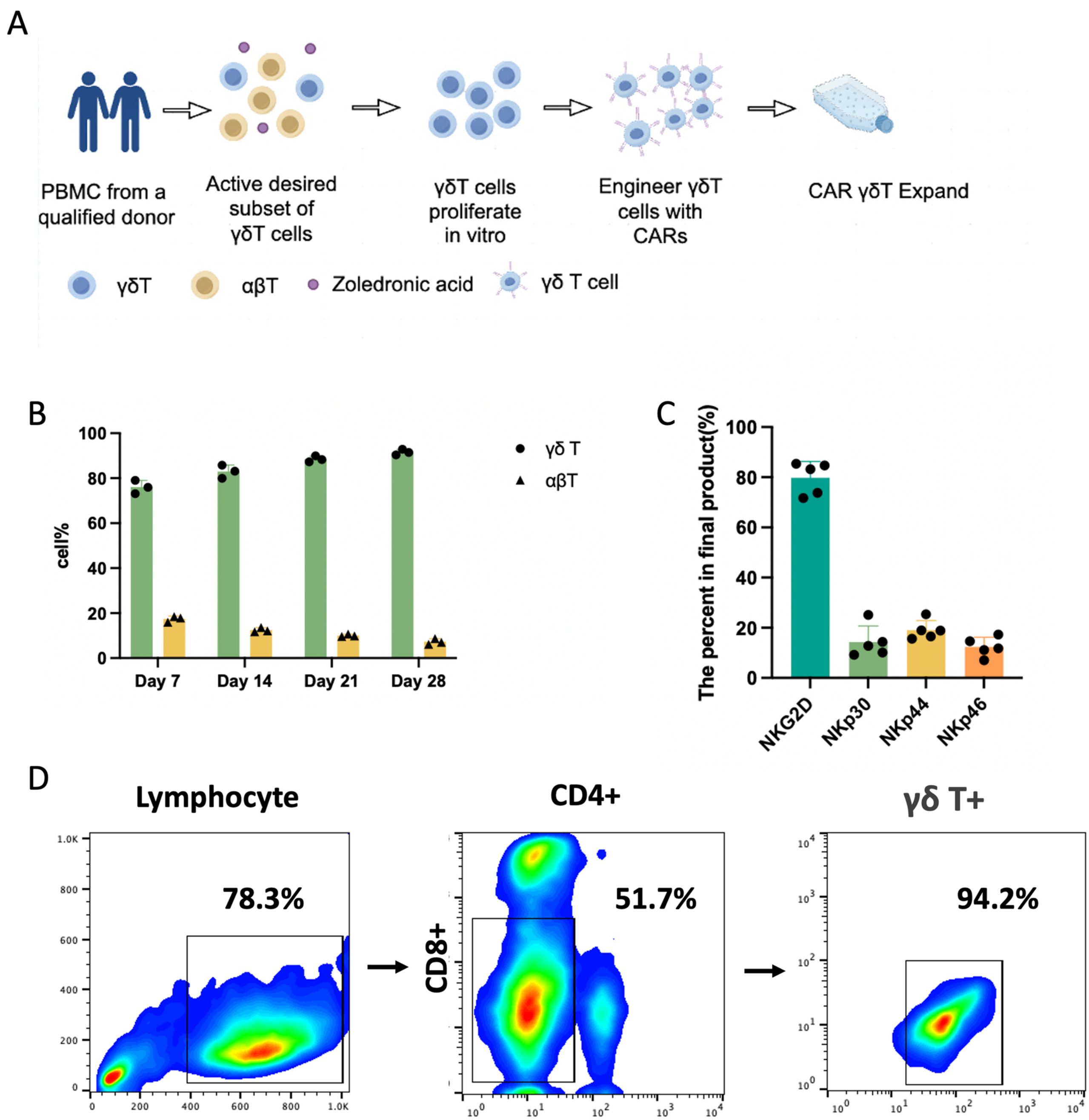

Fig. 2 In vitro expansion and characterization of γδ T cells. The protocol successfully enables the specific expansion of γδ T cells with a phenotype indicative of pre-activated effector potential. (**A**) Schematic of the overall expansion workflow. (**B**) Dynamic changes in the αβ to γδ T cell ratio over time in culture. (**C**) Expression of key surface markers (NK receptors and activation markers) on expanded γδ T cells. (**D**) Final purity (TCRγδ^{+}) of the expanded cell product

of engineering γδ T cells with a CAR, we compared the transduction efficiency of conventional αβ T cells and γδ T cells. Both cell types were transduced with a lentiviral vector encoding a CAR targeting B7–H3 (B7–H3 CAR), and the CAR-positive (CAR^{+}) rate was assessed by flow cytometry after standard in vitro culture. CAR transduction efficiency in αβ T cells was significantly higher than that in γδ T cells (Fig. 3C, D).

To assess the impact of CAR engineering on the expansion capacity of different T cell subsets, we monitored the proliferation kinetics of non-transduced (control T), B7–H3 CAR-αβ T, and B7–H3 CAR-γδ T cells over time, which yielded distinct proliferation profiles for the three groups. Control T cells showed a modest increase in cell number, representing baseline proliferative capacity. Both B7–H3 CAR-AB T cells and B7–H3 CAR-γδ T

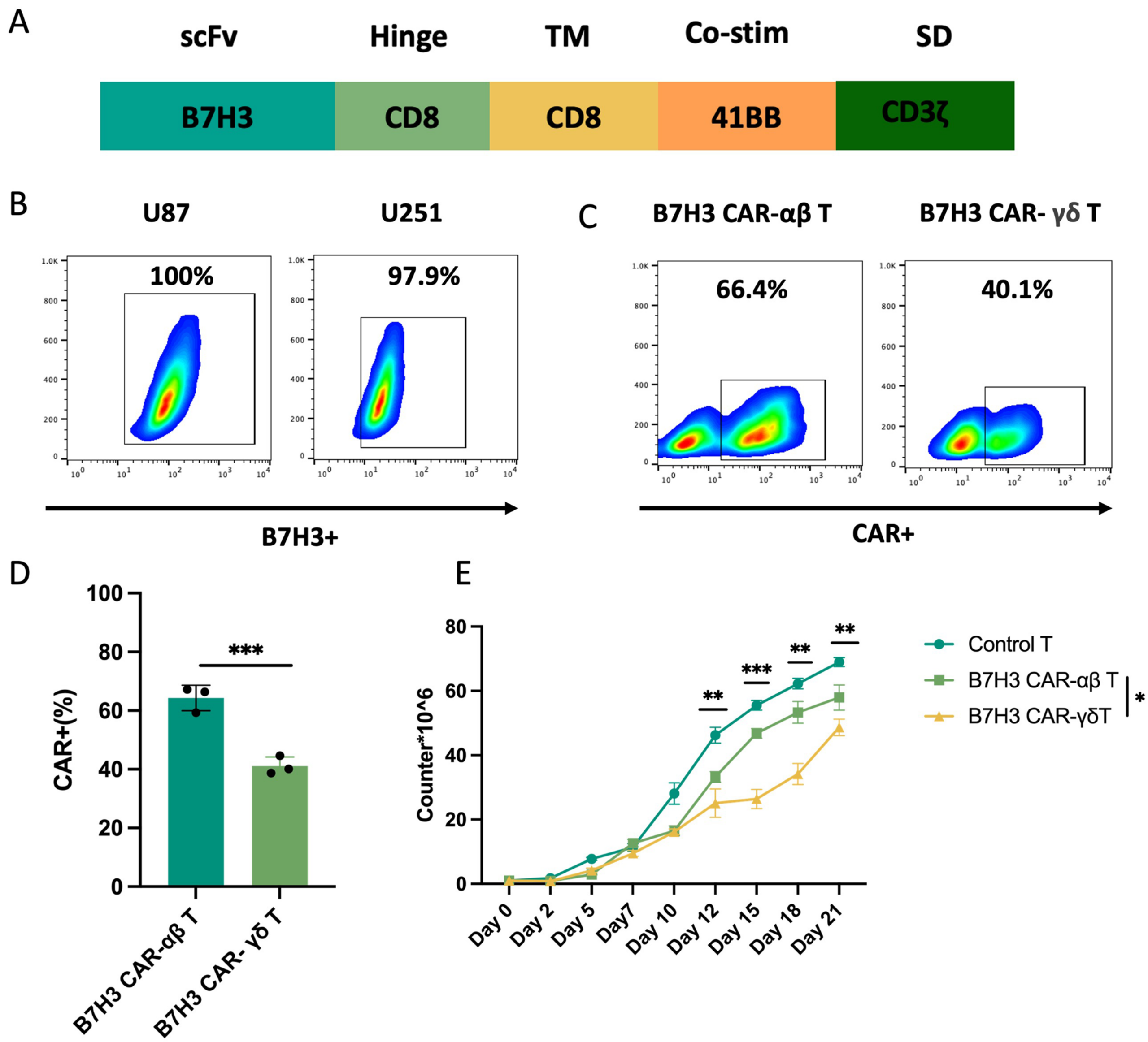

Fig. 3 Generation and characterization of B7–H3 CAR-αβ T cells and B7–H3 CAR-γδ T cells. (**A**) Schematic of the construction of B7–H3-specific CARs. (**B**) B7–H3 was expressed in two glioma cell lines. (**C**) The expression of B7–H3 on the cell surface of two cell lines. (**D**) CAR expression on T cells transduced with B7–H3 CAR-αβ T cells and B7–H3 CAR-γδ T cells on day 7. (**E**) Expansion of T cells transduced with B7–H3 CAR in CAR-T and CAR-γδ T cells. Data shown are mean ± SD ($n = 3$ represents biological replicates from three independent healthy donors). **$p < 0.01$ and ***$p < 0.001$ (two-way ANOVA)

cells exhibited sustained proliferative capacity. Throughout the culture period, the expansion curve of the B7–H3 CAR-αβ T cells was significantly higher than that of the control T cells and the B7–H3 CAR-γδ T cells (Fig. 3E).

Enhanced anti-tumor cytotoxicity of B7–H3 CAR-γδ T cells in vitro

We used U87 and U251 glioma cells to evaluate the activation and cytotoxicity of CAR-T cells against B7–H3-expressing glioma cells. Control cells, B7–H3 CAR-αβ T cells, and B7–H3 CAR-γδ T cells were incubated with the U87 and U251 cells at various effector-to-target (E/T) ratios. Both B7–H3 CAR-αβ T cells and B7–H3 CAR-γδ T cells exhibited U87 cell cytotoxicity, which was positively correlated with the E/T ratio. Cells were quantified based on the number of CAR-positive T cells (Fig. 4A). Notably, B7–H3 CAR-γδ T cells demonstrated greater cytotoxicity than B7–H3 CAR-αβ T cells.

To evaluate the sustainability of CAR-T cell function under repeated antigen exposure, we employed a long-term in vitro tumor challenge model. B7–H3 CAR-T cells and B7–H3 CAR-γδ T cells were co-cultured with U87 cells at an initial 8:1 E/T ratio without cytokine support, with fresh U87 cells added to the co-culture system every 2 days. Quantification of LDH release every 48 hours revealed that B7–H3 CAR-γδ T cells maintained potent

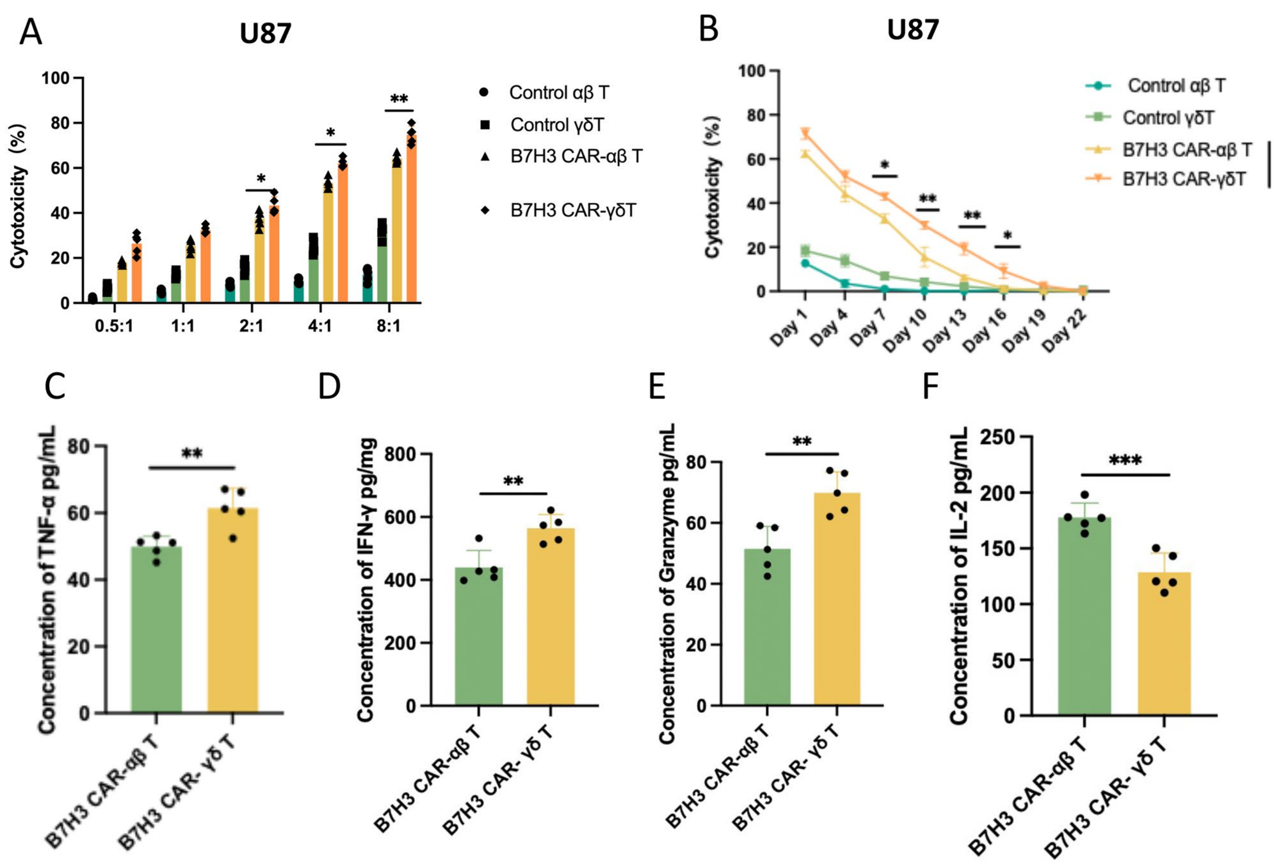

Fig. 4 B7–H3 CAR-γδ T cells enhance anti-tumor cytotoxicity in vitro. (**A**) B7–H3 CAR-αβ T cells and B7–H3 CAR-γδ T cells were co-cultured with U87 cells at different E/T ratios (from 0.5:1 to 8:1). Cytotoxicity was measured using the LDH release assay after 20 h of incubation. (**B**) B7–H3 CAR-αβ T cells and B7–H3 CAR-γδ T cells were co-cultured with multiple rounds of U87 cells (E/T ratios 4:1). Long-term cytotoxicity was measured by the LDH release assay. (**C**-**F**) ELISA data showing cytokine quantification (TNF-α, IFN-γ, granzyme, and IL-2) in the supernatants after B7–H3 CAR-αβ T cells and B7–H3 CAR-γδ T cells were co-cultured with U87 cells at an E:T ratio of 8:1 for 20 h. Data shown are mean ± SD ($n = 5$). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ for the comparison between B7–H3 CAR-αβ and B7–H3 CAR-γδ T cells (two-way ANOVA with Sidak's multiple comparisons test)

tumor-killing activity over multiple rounds of challenge, whereas the cytotoxicity of B7–H3 CAR-T cells decreased more rapidly over time (Fig. 4B). Incubation of the two types of CAR-T cells with U87 cells at an 8:1 ratio also resulted in the release of proinflammatory IFN-γ, TNF-α, IL-2, and granzyme, with B7–H3 CAR-γδ T cells secreting significantly higher levels (Fig. 4C–F).

To investigate whether this sustained functional capacity was associated with a distinct exhaustion profile, we recovered CAR-T cells after the prolonged co-culture and analyzed the expression of key exhaustion markers by flow cytometry. Following this extended antigen exposure, B7–H3 CAR-γδ T cells expressed significantly less PD-1 and LAG-3 than B7–H3 CAR-αβ T cells in both U87 and U251 glioma cells (Supplementary Figure S3). This direct in vitro evidence demonstrates that the superior sustained cytotoxicity of CAR-γδ T cells associates with an inherent resistance to phenotypic exhaustion.

Consistent with these findings, B7–H3 CAR-γδ T cells were significantly more cytotoxic and promoted the release of proinflammatory cytokines when incubated with U251 cells (Supplementary Figure S2).

In vivo efficacy of B7–H3 CAR-γδ T cells

To examine anti-tumor activity of B7–H3 CAR-γδ T cells in vivo, U87 cells were injected subcutaneously into immunodeficient NSG mice to establish a glioma mouse model ($n = 5$). Seven days after U87 cell injection, each group was injected once more with 1×10^7 control or CAR-T cells (Fig. 5A). Tumor-bearing mice were treated with a single dose of control T, B7–H3 CAR-αβ T, or B7–H3 CAR-γδ T cells, and tumor growth was monitored over time. As shown in Fig. 5B, C (U87 model) and Fig. 5D, E (U251 model), both B7–H3 CAR-αβ T and B7–H3 CAR-γδ T cell treatments significantly suppressed tumor growth compared to control, demonstrating specific antitumor activity for B7–H3-targeting CARs. Notably, the antitumor effect of B7–H3 CAR-γδ T cells was significantly higher than that of the B7–H3 CAR-αβ T cells in both models. The kinetics of tumor growth in

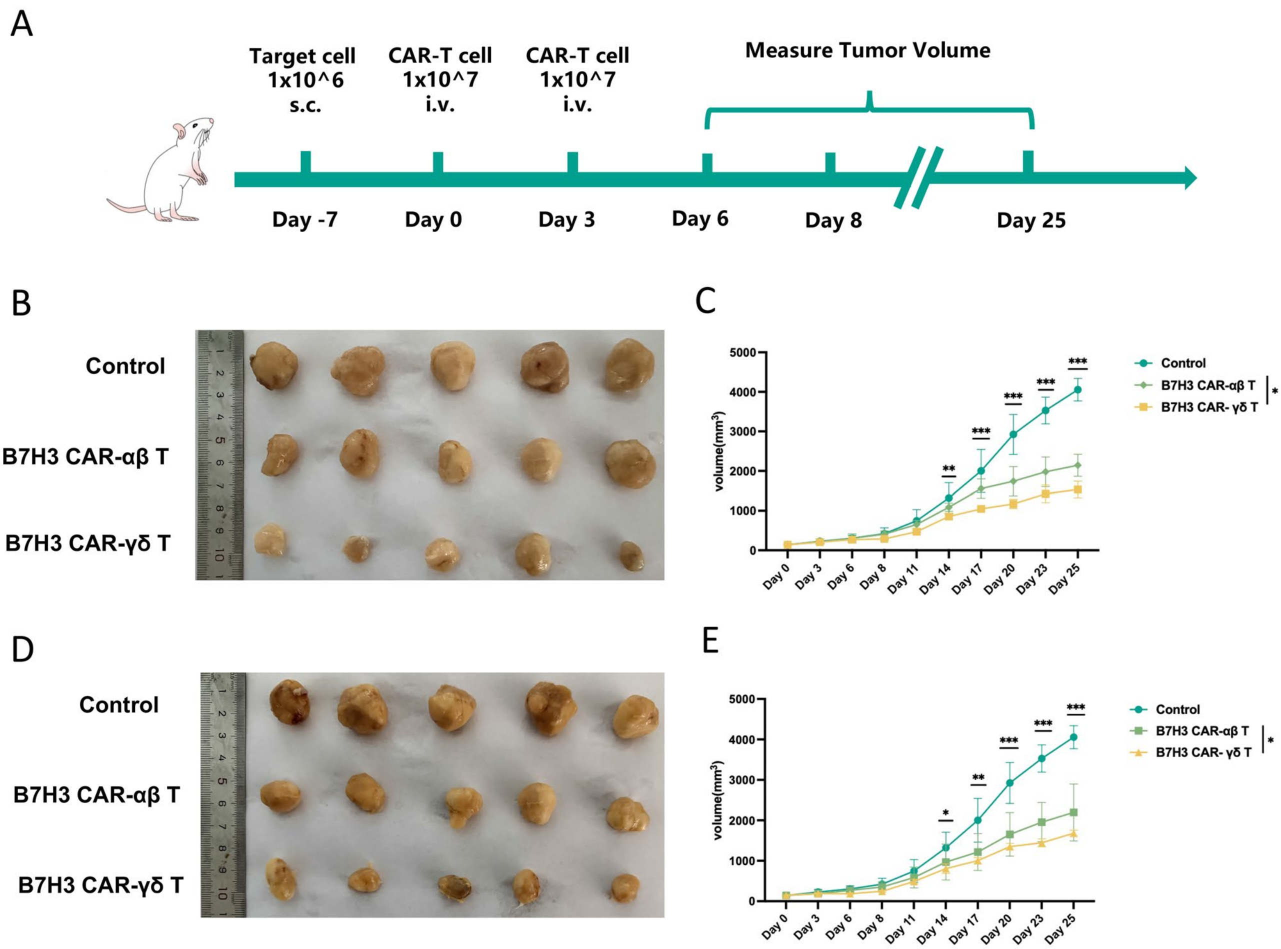

Fig. 5 B7–H3 CAR-γδ T cells show superior anti-tumor activity to B7–H3 CAR-αβ T cells in vivo. (**A**) Schematic of the animal study. (**B**, **C**) Differences in subcutaneous tumor growth in control, B7–H3 CAR-αβ T cell, or B7–H3 CAR-γδ T cell-treated U87 mice. (**D**, **E**) Differences in subcutaneous tumor growth in control, B7–H3 CAR-αβ T cell, or B7–H3 CAR-γδ T cell-treated U251 mice. Data shown are mean ± SD ($n = 5$ mice per group). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ for the comparison between B7–H3 CAR-αβ and B7–H3 CAR-γδ T cells (two-way ANOVA with Sidak's multiple comparisons test)

mice receiving B7–H3 CAR-γδ T cells were significantly slower, resulting in substantially smaller final tumor volumes. This consistent and superior efficacy across two independent glioma models highlights the potent and broad antitumor capability of B7–H3 CAR-γδ T cells in vivo. Toxicity analysis by H&E staining of major organs in B7–H3 CAR-αβ T cells or B7–H3 CAR-γδ T-treated mice revealed no overt histopathological changes, indicating good in vivo biocompatibility.

To explore potential underlying processes contributing to the superior antitumor activity of B7–H3 CAR-γδ T cells in vivo, we performed immunohistochemical analysis of harvested tumor tissues. First, we assessed the tumor-infiltrating capacity of adoptively transferred cells. Immunofluorescence staining revealed significantly greater infiltration of B7–H3 CAR-γδ T cells within the TME compared with B7–H3 CAR-αβ T cells (Fig. 6A, B), suggesting that the enhanced efficacy of B7–H3 CAR-γδ T cells is associated with improved migratory or tumor-homing capabilities. Evaluating tumor cell apoptosis by staining for active caspase-3, tumors from the B7–H3 CAR-γδ T treatment group exhibited significantly higher active caspase-3 expression than those from other groups (Fig. 6C, D). Furthermore, immune checkpoint expression in B7–H3 CAR-γδ T cells was lower than that in B7–H3 CAR-αβ T cells (Fig. 6E–H). Collectively, these data demonstrate that B7–H3 CAR-γδ T cells not only infiltrate tumors more effectively but also induce apoptosis in target cells and reduce immunosuppression, providing a potential mechanism for their superior control of tumor growth in vivo.

Discussion

In this study, we conducted a systematic, side-by-side functional comparison of B7–H3-redirected CAR-γδ T cells and conventional CAR-αβ T cells in the context of glioblastoma. Our results consistently demonstrated that when engineered with the same CAR construct, γδ

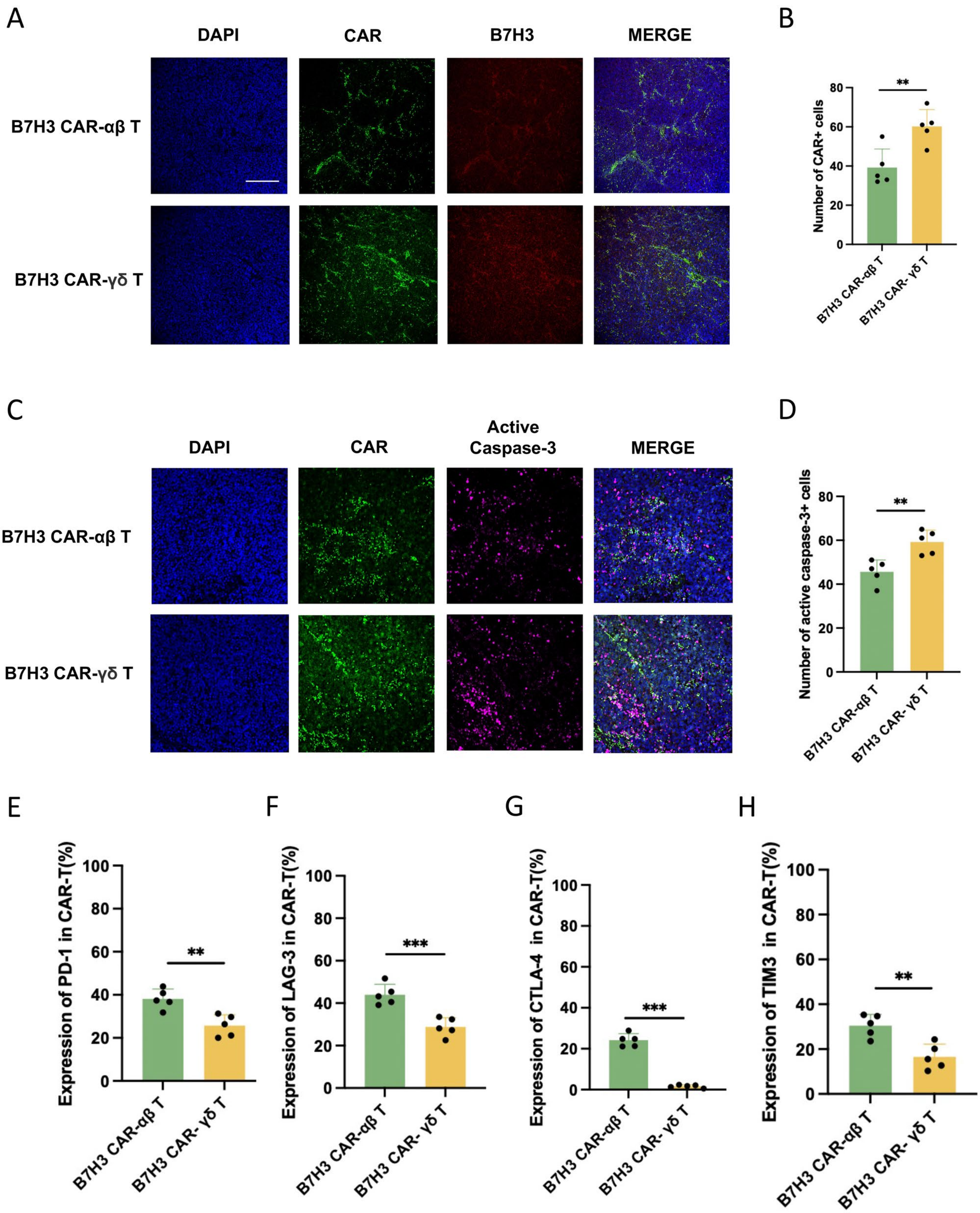

Fig. 6 The potential mechanism associated with B7–H3 CAR-γδ T cell toxicity. (**A**) B7–H3 CAR-γδ T and B7–H3 CAR-αβ T cells infiltrating into tumors were detected using anti-GFP monoclonal antibodies in the U87 model. (**B**) Quantitative analysis of EGFP$^+$ cells in U87 tumor tissues. (**C**) Expression of active caspase-3 was detected by immunofluorescence in U87 tissues. (**D**) Quantitative analysis of active caspase-3 in U87 tumor tissues. (**E**-**H**) B7–H3 CAR$^+$ T cells were stained with antibodies targeting PD-1, CTLA-4, and LAG-3, respectively in U87 tissues. Data shown are mean ± SD ($n = 5$). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ for the comparison between B7–H3 CAR-αβ and B7–H3 CAR-γδ T cells (two-way ANOVA with Sidak's multiple comparisons test)

T cells exhibit markedly enhanced antitumor capacity when examined in multiple assays, including cytotoxicity, cytokine secretion, tumor infiltration, and resistance to exhaustion. These findings provide strong experimental support for the hypothesis that γδ T cells are not only an alternative cellular vehicle for CAR therapy but are also a functionally superior platform for targeting solid tumors such as GBM. Specifically, B7–H3 CAR-γδ T cells not only lysed target cells more effectively across a range of effector-to-target ratios but also maintained sustained killing capacity upon repeated tumor challenge, indicating robust functional persistence. This functional profile is consistent with their polyfunctional cytokine profile of significantly increased secretion of proinflammatory IFN-γ, TNF-α, and granzyme B, which collectively contribute to a more inflammatory and cytotoxic TME.

We chose B7–H3 as the antigen for several clinically-relevant reasons. Unlike other GBM-associated antigens such as EGFRvIII, which is expressed in only a subset of patients and often exhibits heterogeneous and dynamic expression patterns [43, 44], or IL13Rα2, which can be downregulated under immune pressure [8, 45], B7–H3 shows broad, stable, and high expression across the majority of GBMs [17, 21, 46]. This expression profile, validated both through our transcriptomic analysis of TCGA/CGGA cohorts and corroborated by prior immunohistochemical studies, reduces the risk of antigen escape and extends potential applicability to a wider population. Importantly, its limited expression in normal tissues reduces the theoretical risk of on-target, off-tumor toxicity, a critical consideration for clinical translation [47]. By using B7–H3 consistently in our comparisons, we were able to isolate and highlight the intrinsic advantages of the γδ T cell platform rather than conflating them with antigen-specific variables.

Our comparative analyses suggest several functional advantages of CAR-γδ T cells. First, in vivo immunofluorescence analysis revealed that B7–H3 CAR-γδ T cells infiltrated tumor tissues more extensively than their αβ counterparts. Second, these infiltrating CAR-γδ T cells exhibited significantly lower expression of the exhaustion markers PD-1, LAG-3, and CTLA-4. Third, this phenotypic resilience correlated with sustained cytokine production and cytotoxicity in our long-term, multi-round co-culture assays. While these findings clearly demonstrate superior infiltration and a more sustained functional profile, further exploration of the underlying mechanisms is needed. Consistent with established literature, we hypothesize that the enhanced infiltration may be facilitated by the innate expression of specific homing receptors on γδ T cells [48, 49], where CAR signaling may synergize with signaling from natural cytotoxicity receptors such as NKG2D [50, 51]. Similarly, the relative resistance of γδ T cells to exhaustion may be because γδ T cells are better able to withstand immunosuppressive signals within the TME [52]. Importantly, their MHC-independent recognition pathways, which allow them to target stress-induced ligands, may enable CAR-γδ T cells to maintain activity even in immunologically "cold" or antigenically heterogeneous tumors, a common immune evasion strategy in GBM [53–55].

This study has several limitations. First, all in vivo efficacy assessments were in subcutaneous xenograft models, which do not recapitulate critical physiological barriers specific to intracranial GBM, such as the blood-brain barrier and the unique brain immune microenvironment. Second, in vitro studies relied on established glioma cell lines; testing against patient-derived GBM cells, organoids, or slice cultures would strengthen translational relevance and better capture inter-patient heterogeneity, particularly with respect to B7–H3 expression levels and antigen-density–dependent activity. Third, inherent differences in CAR transduction efficiency and expansion kinetics between αβ and γδ T cells, despite normalization in our analyses, suggest that future work should optimize dosing strategies for more precise platform comparisons. Finally, while we observed lower expression of exhaustion-associated markers on CAR-γδ T cells, this remains primarily a phenotypic observation. The functional durability and precise mechanisms underlying this resilience require further investigation through longitudinal stimulation, metabolic profiling, and molecular tracing.

Our work has several translational implications. The intrinsic biology of γδ T cells may enable effective pairing with immune checkpoint blockade, consistent with their lower expression of exhaustion-associated markers. In addition, their capacity for dual recognition via the CAR and endogenous receptors such as the γδ TCR and/or NKG2D could be leveraged through dual-targeting CAR designs or in combination with approaches that increase tumor stress-ligand expression. Finally, γδ T cells may be well suited for allogeneic "off-the-shelf" development because they are not MHC restricted and are generally associated with a lower risk of graft-versus-host disease.

In conclusion, here we provide preclinical evidence that B7–H3 CAR-γδ T cells achieve stronger antitumor activity than matched CAR-αβ T cells in glioblastoma models. Our data support γδ T cells as an infiltrative chassis with a favorable exhaustion-marker profile for CAR engineering. Although additional validation in clinically representative models is now needed, these findings support continued development of γδ T cell-based CAR therapies for GBM and potentially other solid tumors.

Abbreviations

B7-H3	B7 homolog 3 (CD276)
CAR	Chimeric antigen receptor
CAR-γδ T cells	γδ T cells engineered with a chimeric antigen receptor

CAR-αβ T cells	Conventional αβ T cells engineered with a chimeric antigen receptor
GBM	Glioblastoma
TME	Tumor microenvironment
TCGA	The Cancer Genome Atlas
CGGA	Chinese Glioma Genome Atlas
PBMCs	Peripheral blood mononuclear cells
RNA-seq	RNA sequencing
scRNA-seq	Single-cell RNA sequencing
DEGs	Differentially expressed genes
GSVA	Gene set variation analysis
LDH	Lactate dehydrogenase
E/T ratio	Effector-to-target ratio
IFN-γ	Interferon gamma
TNF-α	Tumor necrosis factor alpha
NSG mice	NOD/SCID IL2Rγc-deficient mice
NKG2D	Natural killer group 2 member D
NKp30/44/46	Natural cytotoxicity receptors p30, p44, and p46
PD-1	Programmed cell death protein 1
CTLA-4	Cytotoxic T-lymphocyte–associated protein 4
LAG-3	Lymphocyte activation gene 3

Supplementary information

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Supplementary Material 1

Supplementary Material 2

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Author contributions
Huantong Wu, Tingting Chen,Yajie yu and Shengtao Zhu contributed to the study conception and design. Huantong Wu, Tingting Chen,Yajie Yu perform experiment and analyzed data. Huantong Wu, and Shengtao Zhu wrote and revised the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data availability
The data that support the findings of this study are available within the article and its supplementary materials. Additional data related to this paper are available from the corresponding author upon reasonable request，in accordance with this journal's Share Upon Reasonable Request data policy.

Declarations

Ethics approval and consent to participate
Approval for human sample use by the Ethics Committee of Xuanwu Hospital, Capital Medical University (approval number: [2021]021), along with informed consent.Approval for animal experiments by the Institutional Animal Care and Use Committee of Beijing Friendship Hospital, Capital Medical University protocol number: （GB/T 35,892–2018）, and the animal license number (SYXK 2024–0025) for the mice used.

Consent for publication
All authors approved the final manuscript and the submission to this journal.

Competing interests
The authors have no conflicts of interest to disclose.

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